



FireNet Operations

ACE Team

Aviatrix Transit Firewall Network (FireNet)



Scale out, multi-AZ FW deployments, bootstrapping



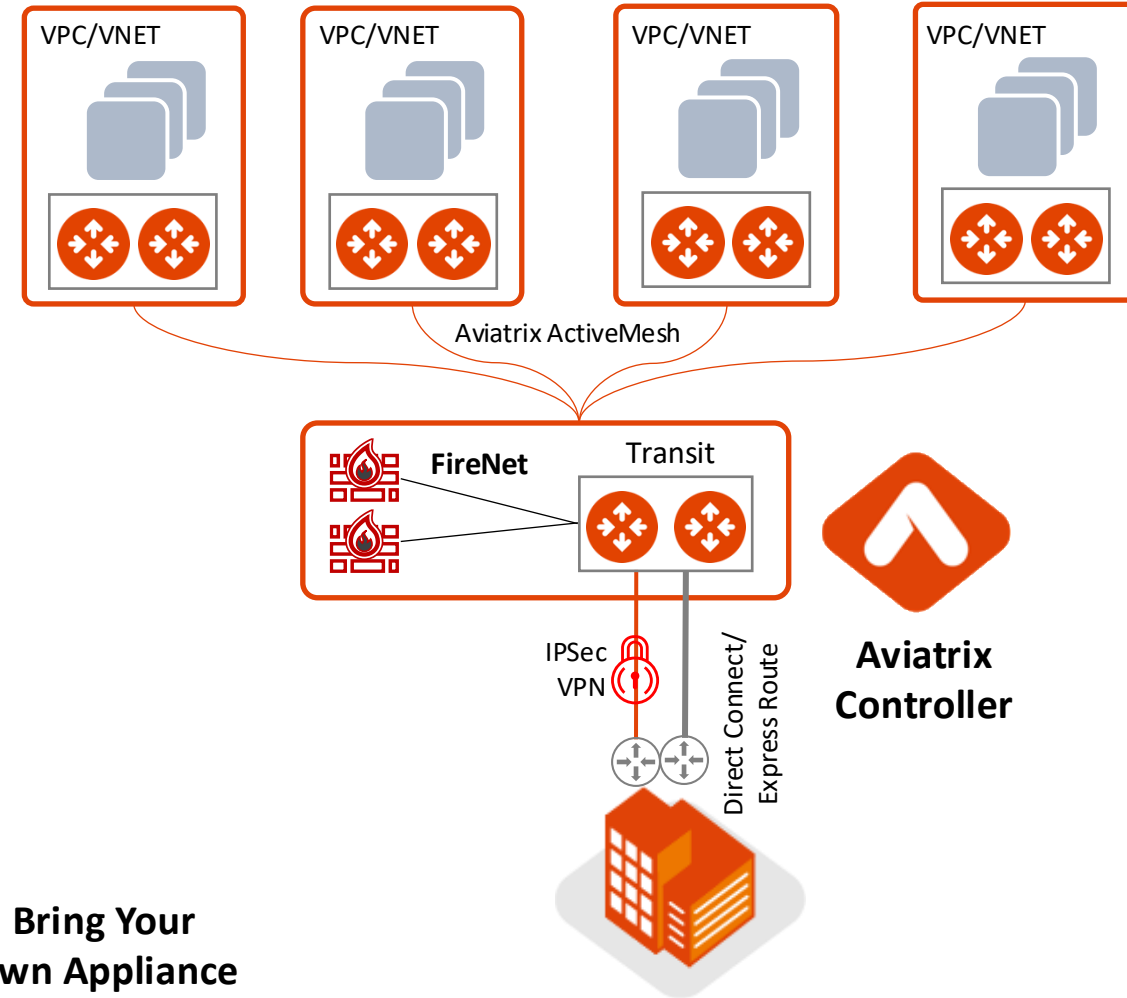
Automated route management, segmentation, and security policies



Deep visibility and operational capabilities



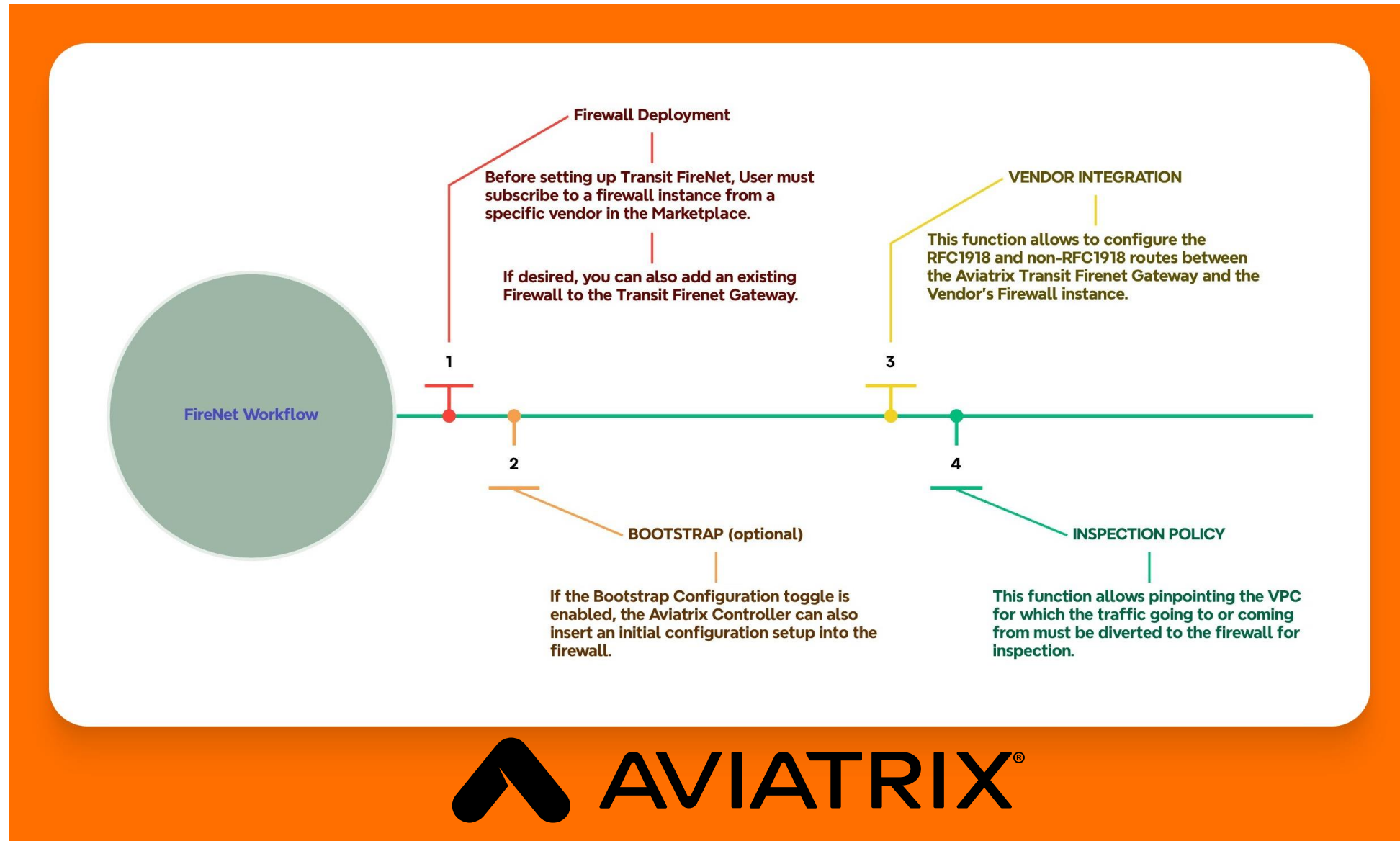
Repeatable across regions and clouds



Bring Your Own Appliance



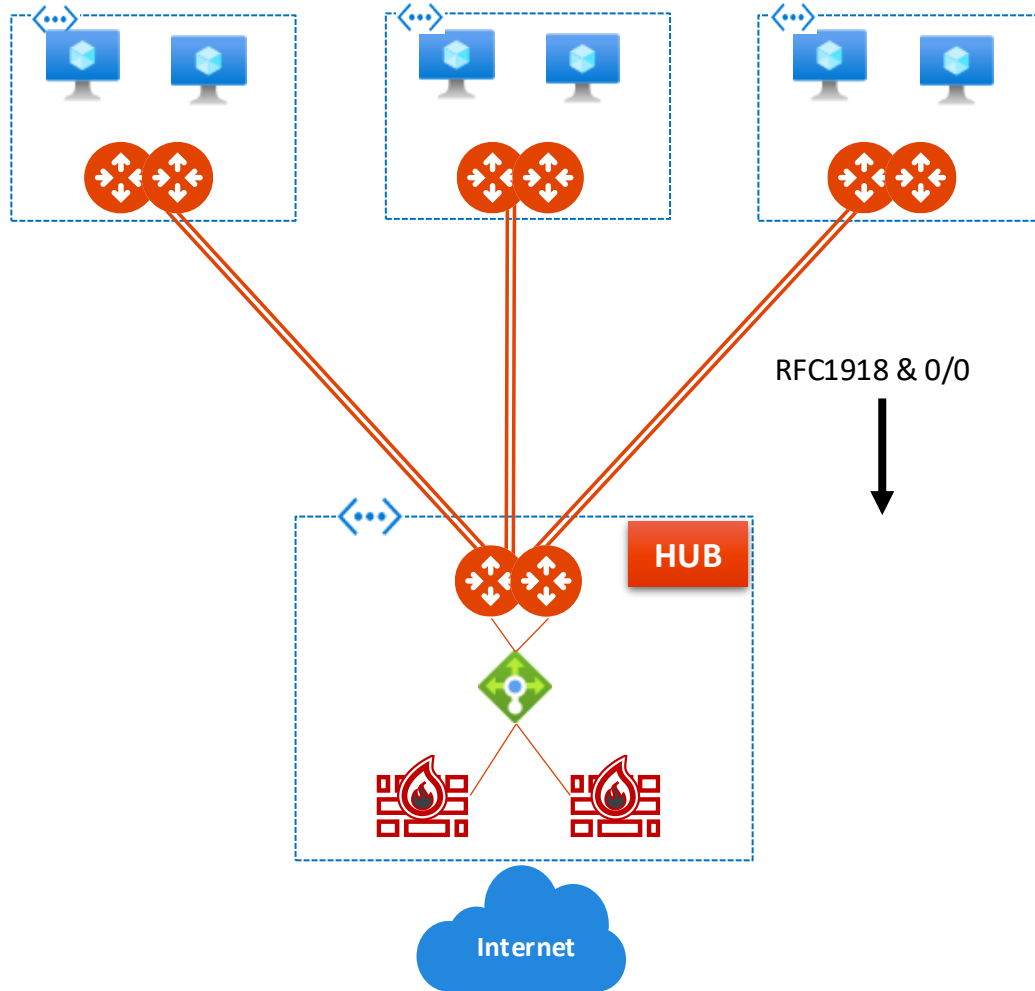
FireNet Deployment



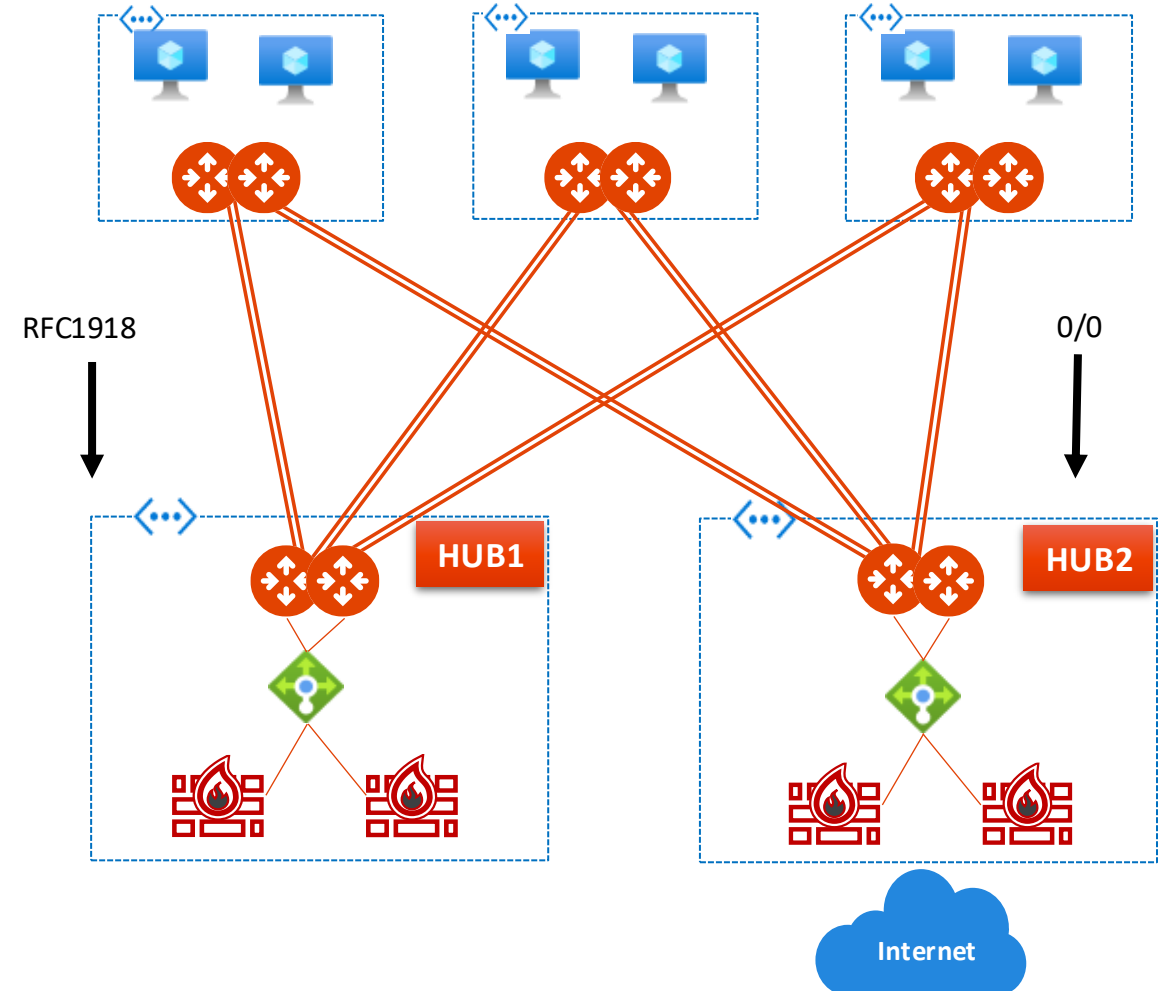
FireNet Architecture Options (Azure Example)

Each firewall set can scale independently based on need

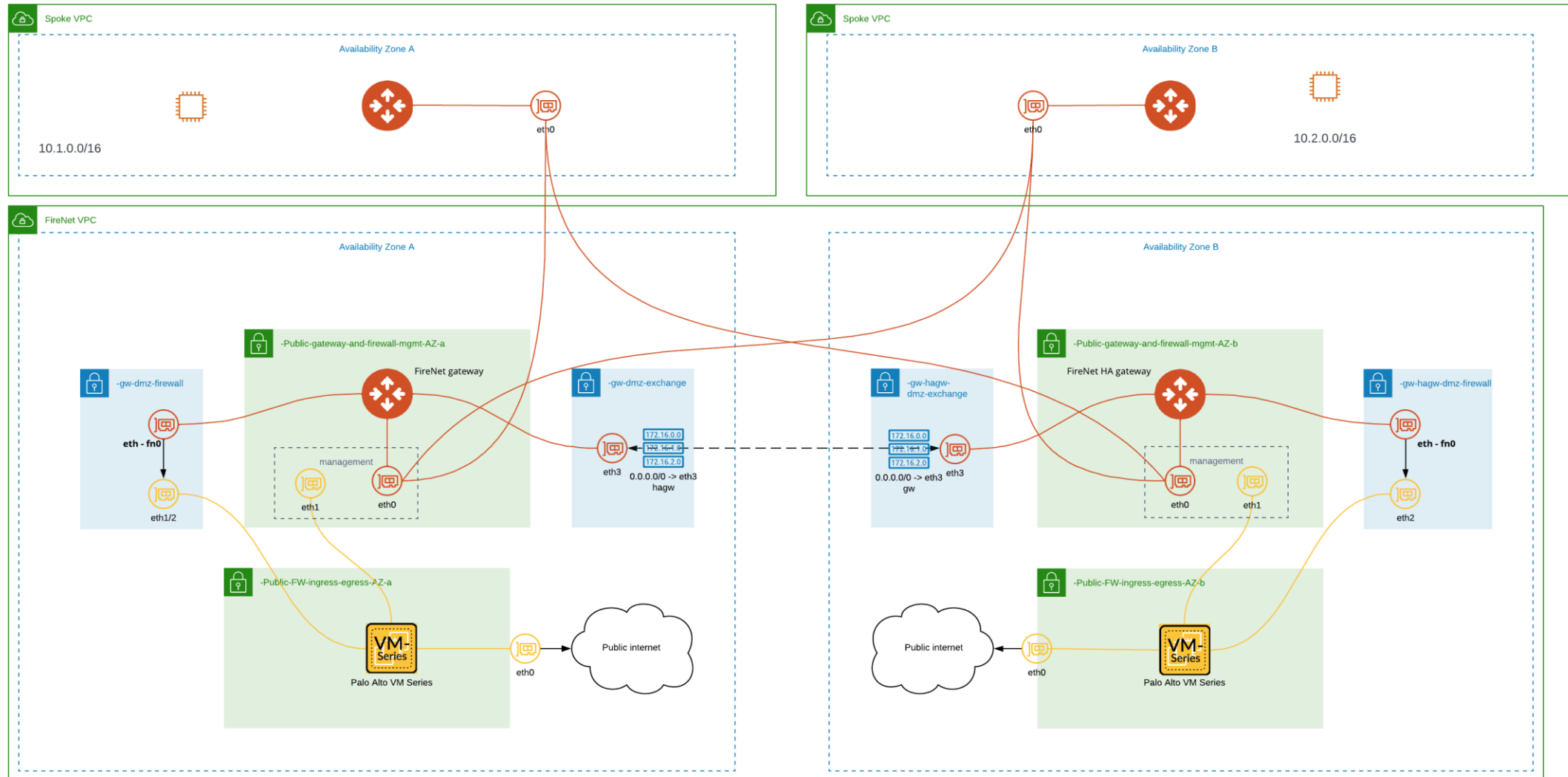
Single HUB FireNet



Dual HUB FireNet



FireNet – Under the hood





Tools for Operating your FireNet

Firewall Deployment Workflow



- **PATH:** Security > FireNet > Firewall

1. Select the Transit FireNet GW
2. Select the Firewall Image (requirement: *Subscribe to the firewall instance from the Marketplace*)
3. Firewall Image Version
4. Firewall Instance Size
5. Egress Interface Subnet
6. Management Interface Subnet (Palo Alto/AWS only)
7. Bootstrap Configuration (*optional*)

- **Supported Firewall Vendors:** Palo Alto VM-Series, Check Point CloudGuard, Fortinet FortiGate, BYOA

- **Panorama** is also supported as a firewall manager for Palo Alto VM-Series.

CoPilot

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Cloud Fabric

Networking

Security

Distributed Cloud Firewall

Egress

ThreatIQ

FireNet

Anomaly Detection

Groups

Cloud Resources

Monitor

Diagnostics

Administration

Settings

FireNet

FireNet Gateways

Firewall

+ Firewall

Name

Vendor

ACE-FW

Fortinet FortiGate

Deploy Firewall

After a firewall instance is launched, wait for 15 minutes for it to come up.

Association

FireNet Instance

ace-aws-eu-west-1-transit1

Attach Firewall to FireNet after Launching

Yes

Firewall Configuration

Name

FORTIGATE-FW

Firewall Image

Fortinet FortiGate Next-Generation Firewall

Firewall Image Version

(7.6.2)

Firewall instance Size

c4.xlarge

Egress Interface Subnet

10.1.200.0/28~~eu-west-1a~~ace-aws-eu-west-1-t...

Key Pair Name

Optional

Bootstrap Configuration

Cancel

Deploy

Bootstrap (*optional*)

- **Botostrap Configuration toggle**

- *Toggle Disabled* (default): the FW is deployed with an empty configuration
- *Toggle Enabled*: the FW is deployed with an initial configuration
 - You need to specify the Location where the AVX Controller will retrieve the initial configuration (e.g. Azure Storage, S3 Bucket, etc.)

☒ Bootstrap Configuration

☒ Azure Storage ☐ User Data

Storage

pod74firenetbootstrap

Storage Access Key

aMef4Dlr3m9wh/T+9o73FPHonOSsVUC+AStlPOrhQ==

File-Share Folder

pod74-firenet-bootstrap

Share-Directory

Optional

Vendor Integration

- The Vendor Integration function allows the Controller to log into a firewall or firewall manager and change the route table on the firewall to program the routing for FireNet, or to change routing if a gateway in FireNet fails.
- Vendor Integration allows to configure the **RFC 1918 routes** and **non-RFC 1918 routes** on the Vendor's firewall instance

Vendor Integration

FireNet

ace-aws-eu-west-1-transit1

☒ Through Firewall
 ☐ Through Firewall Manager

☒ Firewall

ACE-FW

Management IP Address

52.49.101.236

Vendor

Palo Alto Networks VM- ... x v

Username

avxadmin

Password

Route Table

Optional

Revoke Integration

Cancel

Save

ACE-FW

Name: ACE-FW

Vendor: Fortinet FortiGate

Public IP: 54.76.250.245

Static Route Table:

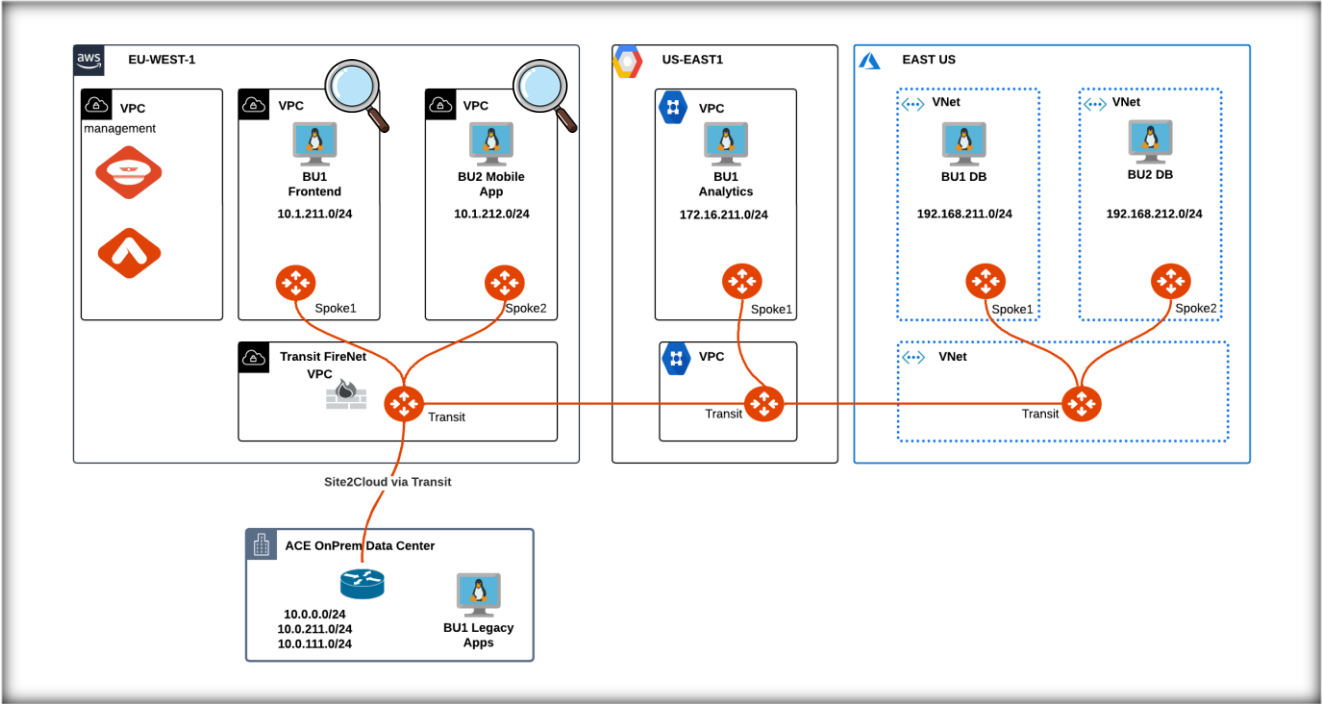
Destination	Gateway IP	Interface	Distance	Weight	Status	Comment
172.16.0.0/12	10.1.200.65	port2	10	0	enable	Aviatrix Vendor Integration
192.168.0.0/16	10.1.200.65	port2	10	0	enable	Aviatrix Vendor Integration
10.0.0.0/8	10.1.200.65	port2	10	0	enable	Aviatrix Vendor Integration

PBR:

Destination	source	In	Intf	protocol	Src Ports	Dst Ports
Gateway	Out Intf	Status	action	Comment		

Inspection Policy

- On the FireNet **Policy** tab you can add or remove the **inspection policy** for the selected VPC/Vnet/VCN. When an inspection policy is added the traffic related to the Transit FireNet's attachment (Spoke/Edge gateway, peered Transit, Site2Cloud connection) is inspected by the firewall within the selected Transit FireNet.
- By default*, FireNet inspects ingress and east-west traffic only.



FireNet FireNet Gateways Firewall

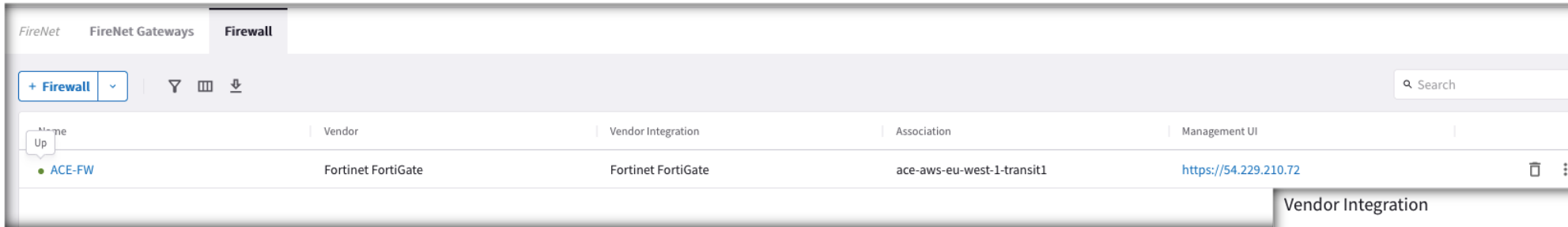
ace-aws-eu-west-1-transit1

Firewall **Policy** VPC/VNet Route Tables Vendor Integration Settings

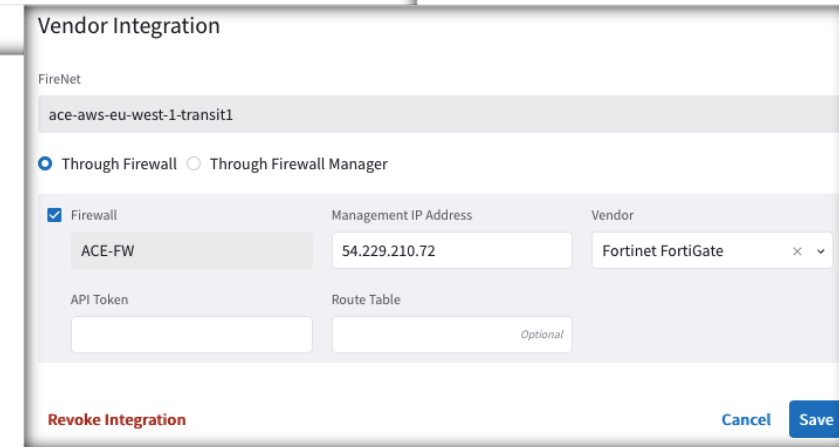
Actions Azure Spoke Subnet Groups

☐ Name	Type ↓	Inspection
☐ ace-aws-eu-west-1-spoke1	SPOKE	On
☐ ace-aws-eu-west-1-spoke2	SPOKE	On
☐ ACE-ONPREM-DC	SITE2CLOUD	Off
☐ ace-gcp-us-east1-transit1	PEERING	Off
☐ ace-azure-east-us-transit1	PEERING	Off

Information to Collect / Initial Troubleshooting



Name	Vendor	Vendor Integration	Association	Management UI
ACE-FW	Fortinet FortiGate	Fortinet FortiGate	ace-aws-eu-west-1-transit1	https://54.229.210.72



Vendor Integration

FireNet

ace-aws-eu-west-1-transit1

☒ Through Firewall ☐ Through Firewall Manager

☒ Firewall

ACE-FW

Management IP Address: 54.229.210.72

Vendor: Fortinet FortiGate

API Token:

Route Table: Optional

Revoke Integration

Cancel Save

- Make sure Aviatrix sees the FW as “healthy”
 - For Ingress: Check if any native LB deployed in front of the FWs is also configured correctly
- Vendor Integration: make sure the controller can reach the FW
 - Nothing preventing the communication, NACLs, NSGs, SLs, etc.
- Make sure there are no “uncommitted” pending changes on the FW
- Make sure your Network Domain/Spoke is configured for inspection
- Make sure *Connected Transit* is enabled (if necessary)
- Make sure your Spoke is attached to Transit
- Verify Spoke and Transit GW routes in Cloud Fabric > Gateways



Next:

Lab 3 FireNet - Interface
Lab 4 FireNet - Routes